

Spatial Analysis of Human Trafficking Vulnerability

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Vulnerability in the Pittsburgh MSA

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Introduction and Context

Our project focuses on human trafficking and how it is a pervasive and hidden crisis affecting vulnerable populations across Western Pennsylvania, some possibilities of these individuals include being homeless, living in a single parent or low-income household, along with many more common characteristics that cause people to be more vulnerable. The project employs Geographic Information Systems (GIS) to spatially analyze risk factors associated with human trafficking in the Pittsburgh metropolitan area. By integrating data on homeless shelters, high-risk trafficking zones, law enforcement locations, and socioeconomic indicators, the study aims to identify spatial patterns that can inform prevention and intervention strategies.

The Crises of Human Trafficking

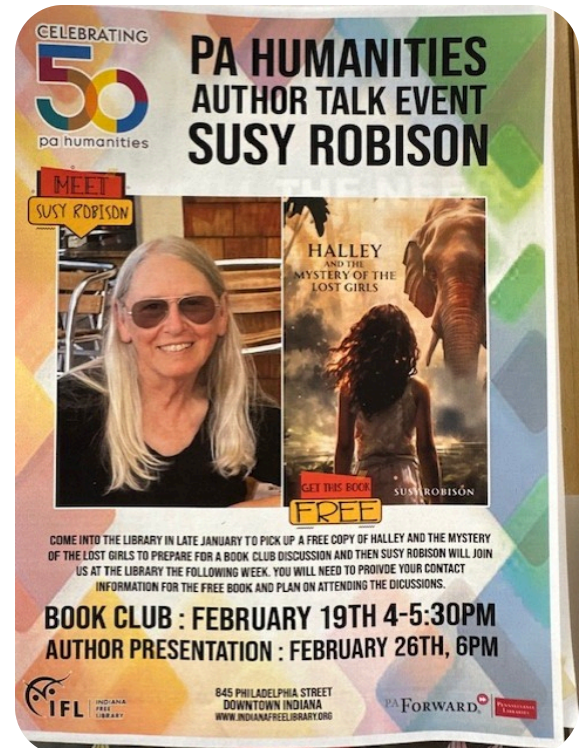
- The United Nations has defined human trafficking as the recruitment, transfer, harboring, or receipt of persons by threat or use of force.
- According to the UN, a trafficker is part of the act of recruiting, transporting, or receiving of a person, through criminal means ranging from brute force and purchase of a person to taking

advantage of someone in a vulnerable situation, and whose purpose is gross exploitation (United Nations, 2004).

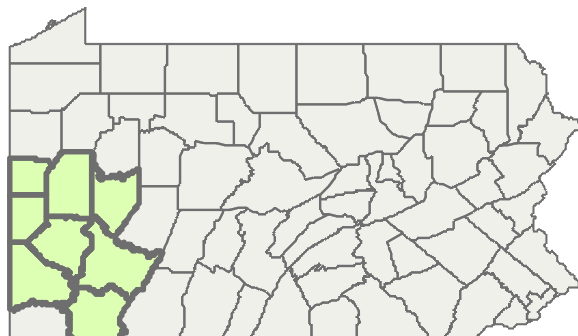
- There are disproportionately higher rates of African American and Latinx child populations in out-of-home placements. Children of color are more likely to be exploited by traffickers. Some studies showing that 50% to 90% of identified trafficking victims having a history of foster care and/or being homeless (Child Welfare Information system, 2017) (Epstein, Blake, J., & González, 2017).
- Studies on homeless youth explain that being homeless increases the likelihood of becoming a victim of sexual exploitation and human trafficking (Choi, 2015) (Casey Family programs, 2014).

Local Resources

We had the privilege of visiting the Indiana Community Library to attend a compelling talk by author Susy Robison. Drawing from her experiences as a volunteer in homeless shelters, Robison became acutely aware of the alarming prevalence of human trafficking within these vulnerable communities. This revelation led her to join an organization in the Pittsburgh metropolitan area dedicated to reducing the trafficking of individuals experiencing homelessness. Her work eventually brought her into collaboration with the FBI, local law enforcement, and a network of advocacy groups—all united in the mission to combat human trafficking. Through these efforts, Robison has made meaningful contributions to addressing and decreasing the incidence of trafficking in at-risk populations.

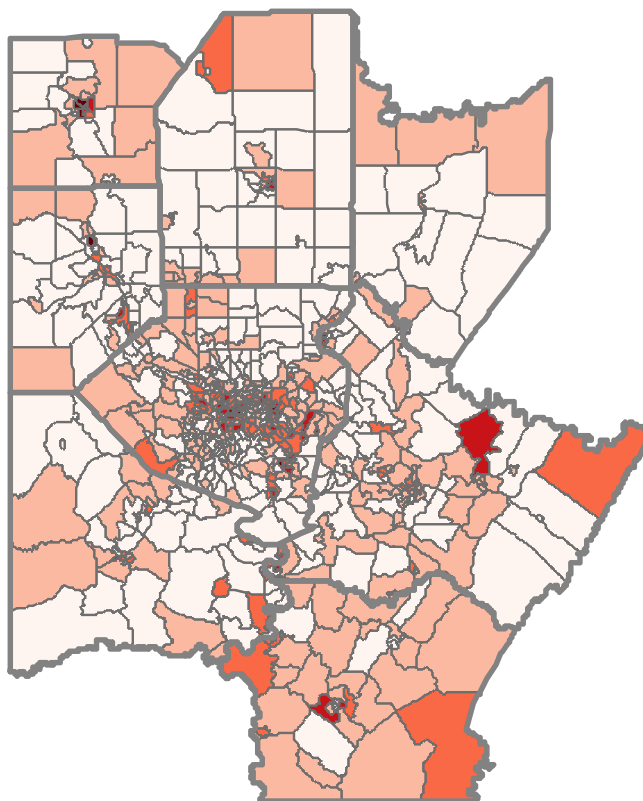


Data Development



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Reference Map of our Project Study Area

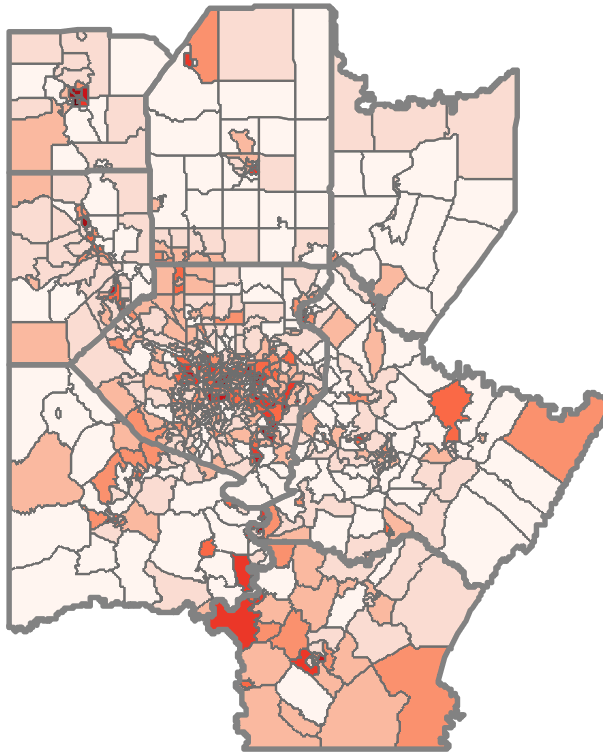


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Pittsburgh MSA County Layer and Index Tract

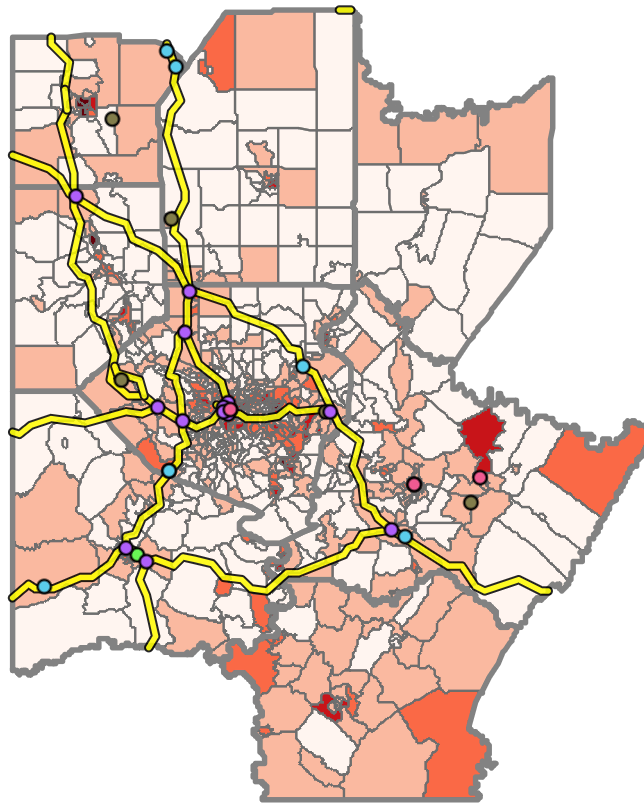
This map presents the human trafficking vulnerability index across census tracts in the Pittsburgh MSA. Tracts are color-coded by grouped index scores (0–1, 2–3, 4–5, 6–7, 8–9), with darker shades indicating higher vulnerability. The index was calculated using demographic, socioeconomic, and infrastructure variables. This map serves as the central output of the project, highlighting areas most at risk.

Our feature class was created by joining two ACS data tables to the census tract feature class using shared geographic identifiers. After the join, statistical analysis was used to identify tracts in the highest quartile of vulnerability across key variables such as income, unemployment, age, gender, household structure, race, non-citizenship, and mobility. Tracts were further analyzed based on proximity to transportation nodes and homeless shelters. Additionally, tracts relying solely on state police coverage were assigned extra vulnerability weight.

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High Vulnerability- Socioeconomic Factors Only

This map isolates tracts with high vulnerability based solely on socioeconomic indicators such as poverty, unemployment, single-parent households, non-citizen populations, and mobility. By excluding infrastructure and location-based variables, the map highlights communities where social and economic instability alone may increase trafficking risk. This targeted view helps differentiate structural vulnerability from geographic exposure.

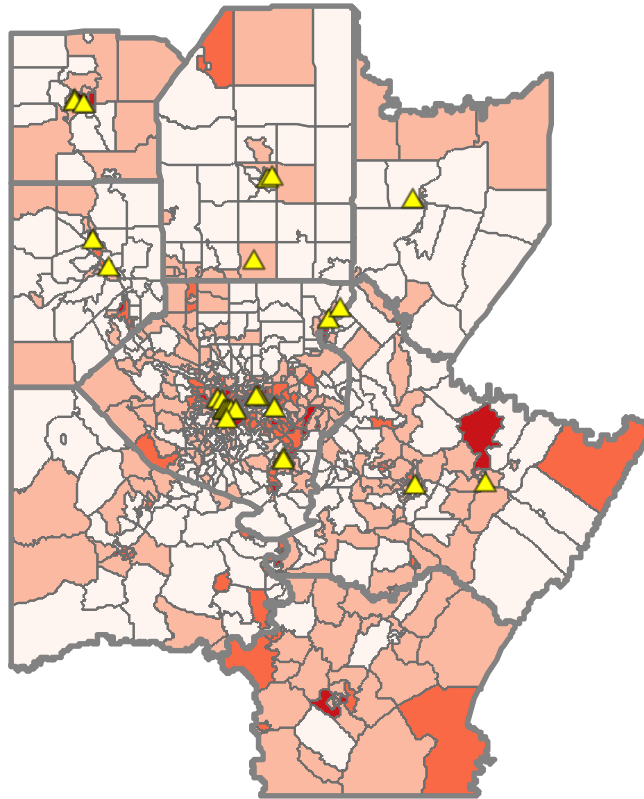


Members of the Geographic Information Systems Laboratory, Pellissippi...

50 km | Powered by Esri

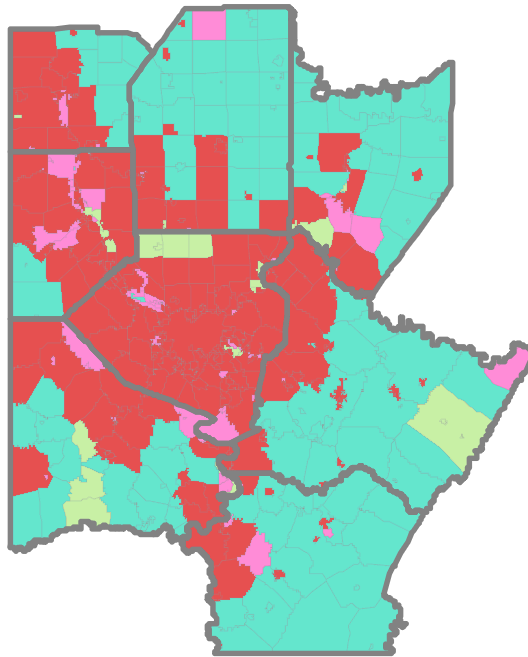
Transportation Layer

This map highlights major transportation features in Pittsburgh MSA, including highways, arterial roads, rail lines, intercity bus terminals (Greyhound, Trailways, Fullington), and truck stops. These elements were mapped to identify high-access areas that may serve as trafficking corridors. Understanding the transportation network is essential for analyzing how mobility and infrastructure contribute to human trafficking vulnerability.

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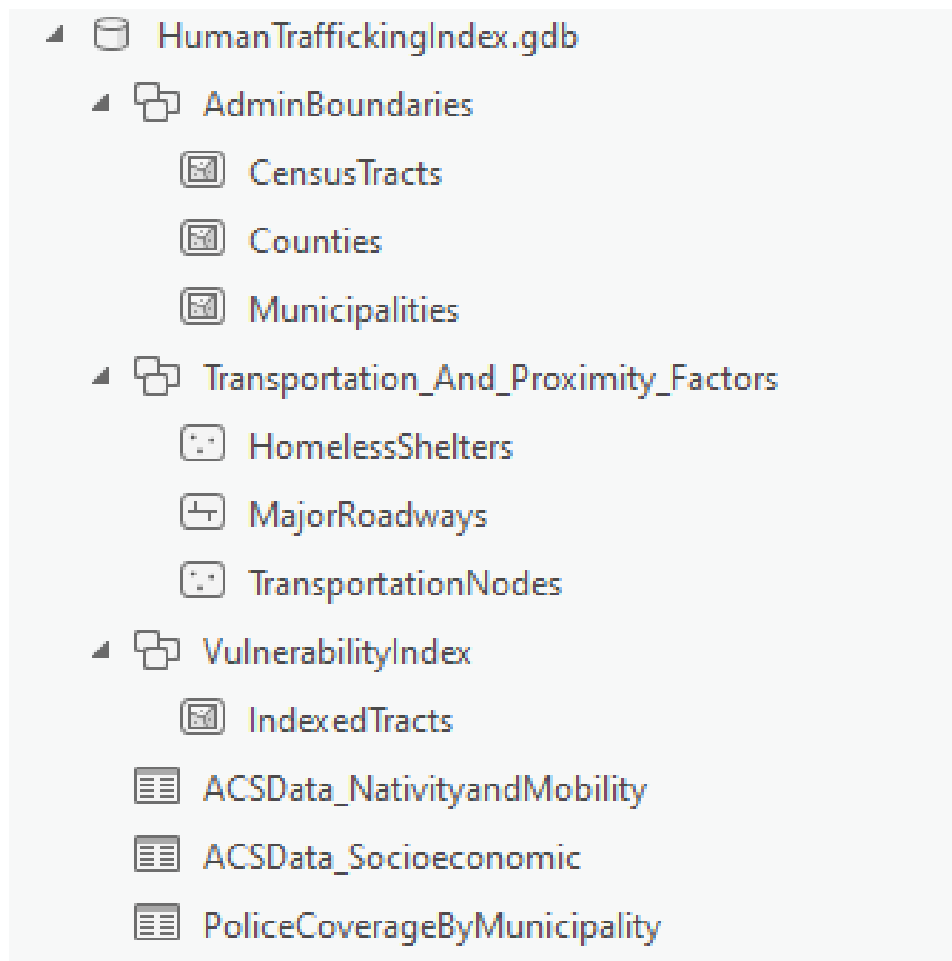
Homeless Shelter Layer

This map shows the locations of homeless shelters throughout Pittsburgh MSA, represented as point features. These sites are important indicators in the analysis, as individuals experiencing homelessness are at higher risk for trafficking due to unstable housing and limited resources. The map helps visualize shelter distribution in relation to other risk factors.

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Police Coverage

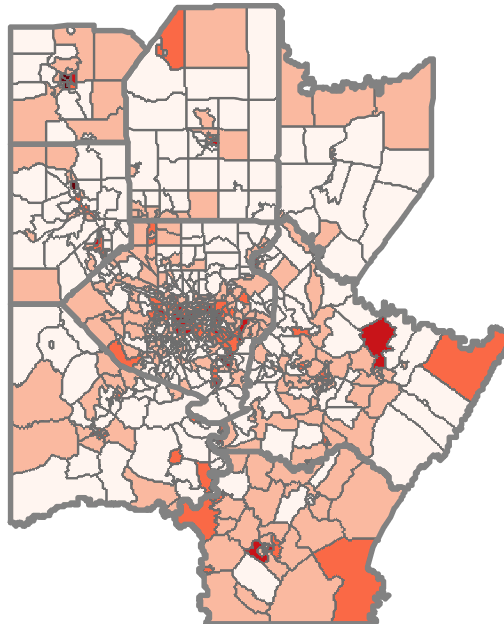
This map displays the locations of police stations across the Pittsburgh MSA and illustrates general coverage areas based on proximity. The purpose is to identify regions with lower access to law enforcement, which may correlate with increased vulnerability to human trafficking. Police station data was mapped as point features and analyzed to highlight potential enforcement gaps.



Geodatabase Structure

Analysis and Results

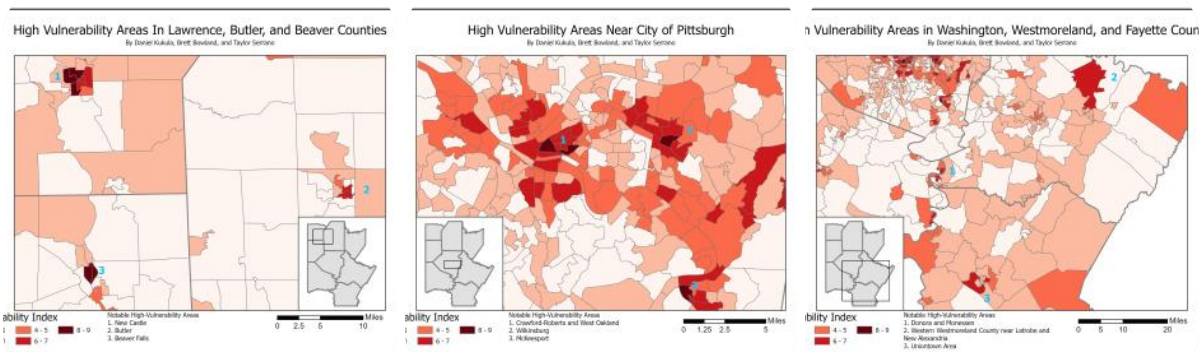
The following Map represents the final result of our analysis.



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Final Indexed Map

The gallery below highlights key areas of observation where individual census tracts or broader areas showed high levels of vulnerability:



Notable Trends

- Many of the most vulnerable areas are historic river-front towns —such as New Castle —that once thrived during the coal and steel era. These "rust-belt" towns have experienced long-term economic decline, which correlates with high index scores due to persistent poverty, unemployment, and population instability.

- Disadvantaged neighborhoods within the Pittsburgh urban core, including Homewood, East Hills, Arlington, Rankin, and McKeesport, also score high on the index. These communities share key risk factors with rural and suburban towns—like poverty, high minority populations, and housing instability—but are further shaped by their proximity to dense transit networks and urban disinvestment.
- Transportation infrastructure (bus stations, truck stops, major roads, and railways) strongly overlaps with areas of high vulnerability, reinforcing the role of transit access in trafficking movement and exposure.
- Homeless shelters and law enforcement presence vary widely across the region. Vulnerable areas with limited police coverage and nearby shelter access—especially outside Allegheny County—may lack critical support or intervention capacity.
- Socioeconomic vulnerability alone (e.g., high poverty, non-citizen population, mobility, and single-parent households) accounts for major portions of the index variation, even in areas with limited transportation exposure.
- Index ranges as high as 6–9 appear in both dense urban neighborhoods and small towns, emphasizing that human trafficking vulnerability is a regional problem, not confined to any one geographic type or location.

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